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Z00501 – MID TERM SOLVED

What is gastrulation?

Gastrulation is a phase early in the embryonic development of most animals, during which the single-layered blastula is reorganized into a multilayered structure known as the gastrula.

Who discover sperm?

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in 1678

Name the three germ layer in animals?

1. Ectoderm 2. Mesoderm 3. Endoderm

How many type of mRNA are present in sea urching egg?

sea urchins contain 25,000 to 50,000 different types of mRNA.

Differentiate B/W Promoter and Enchancer?

Promoter: A region, which is responsible for the binding of RNA polymerase and for the subsequent initiation of transcription.

Enhancer: It is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs

What are Somites?

Somites are precursor populations of cells that give rise to important structures associated with the vertebrate body plan and will eventually differentiate into dermis, skeletal muscle, cartilage, tendons, and vertebrae. ... Formation begins as paraxial mesoderm cells organize into whorls of cells called somitomeres.

What is different B/W Sex and Fertilization?

Fertilization: Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Sex involves the combining of genes from two different individuals into new arrangements.

What are the event of Fertilization?

Although the details of fertilization vary from species to species, conception generally consists of four major events:

- Contact and recognition between sperm and egg.□ In most cases, this ensures that the sperm and egg are of the same species.
- Regulation of sperm entry into the egg. Only one sperm can ultimately fertilize the egg
- Fusion of the genetic material of sperm and egg.
- Activation of egg metabolism to start development.

What are two main startegie of developmental commitment?

There appear to be two major strategies for establishing commitment and hence initiating the series of events that result in cell differentiation.

The inheritance of cytoplasmic determinants: Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely intrinsic

The perception of external inductive signals: The process where one cell or group of cells changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

How Egg is protected form sunlight? Which of Egg protected form sunlight?

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight; some eggs contain molecules that potential predators find distasteful; and the yolk of bird eggs even contains antibodies.

Composition of Egg Yolk? Explain composition of Egg Yolk?

Yolk vitamins and minerals. **Yolks** hold more than 90% of the calcium, iron, phosphorus, zinc, thiamine, vitamin B_6, folate, vitamin B_{12}, and pantothenic acid of the **egg**. In addition, **yolks** cover all of the fat-soluble vitamins: A, D, E, and K in the**egg**, as well as all of the essential fatty acids.

Conditional Specification?

Conditional specification: o Characteristic of all vertebrates and few invertebrates. o Specification by interactions between cells. Relative positions are important. Variable cleavages produce no invariant fate assignments to cells. o Massive cell rearrangements and migrations precede or accompany specification. o Capacity for "regulative" development: allows cells to acquire different functions.

What is difference B/W protocadherin and other Cadherin?

Protocadherins: Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenins. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

E-cadherin: Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1-cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.
P-cadherin: Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.
N-cadherin: Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their Ecadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Describe anatomical approachesof developmental biology?

The basis of all research in developmental biology is the changing anatomy of the organism. What parts of the embryo form the heart? How do the tissues that form the bird wing relate to the tissues that form the fish fin or the human hand? Following important parameters weave together to form the anatomical approaches to development.

Comparative embryology Evolutionary embryology Teratology Mathematical modeling

What is Microvilli? Microfilaments?

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections called microvilli, which may aid sperm entry into the cell.

Difference B/W Somatic Cell and Germ Cell?

Germ Cell: In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.
Somatic Cell: All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development.

How the gene activity of human Is obtain?

Knowledge of gene activity in humans can be obtained by candidate gene mapping or positional cloning.

Transcription Factor Domain?

Transcription factors have three major domains. 1. DNA-binding domain 2. Protein-protein interaction domain 3. Trans-activating domain.

Define adherence Junction?

Adherens junctions (AJs) are cell-cell adhesion complexes that are continuously assembled and disassembled, allowing cells within a tissue to respond to forces, biochemical signals and structural changes in their microenvironment.

What is function of vitelline envelop?

The vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

Define Organogenesis in term of germ layer?

Once the three germ layers are established, the cells interact with one another and rearrange themselves to produce tissues and organs. This process is called organogenesis. Many organs contain cells from more than one germ layer.

Sperm Egg reorganization?

The interactions and fusion of the **sperm** and **egg**, changes are initiated that affect virtually all components of the **egg** cortex. The **egg** plasmalemma becomes a mosaic composed of membranes derived from the **egg** and **sperm**, as well as from dehisced cortical granules.

Development of hierarchy of animals?

The developmental hierarchy is the hierarchical series of decisions involving specific pre-existing cell types. These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.

Structure of Sperm?

The **sperm** head. The head is oval-shaped and is only 10% of the overall length of the **sperm**. It contains the male genetic material in a **structure** called the nucleus. ... In addition to DNA, the **sperm** head also contains a**structure** called the acrosome.

Mosaic development?

Autonomous specification gives rise to a pattern of development referred to as mosaic development, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts.

Capacition?

Capacitation is the penultimate step in the maturation of mammalian spermatozoa and is required to render them competent to fertilize an oocyte. This step is a biochemical event; the sperm move normally and look mature prior to capacitation.

Poly A tale Function? Role of Poly A tale in mRNA?

The poly(A) tail Confers stability on the messenger RNA. Allows the mRNA to exit the nucleus. Permits the mRNA to be translated into protein.

Vegetal Pole?

The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

What is the larve stage of frog?

A tadpole (also called a pollywog) is the **larval stage**in the **life cycle** of an amphibian, particularly that of a**frog** or toad. They are usually wholly aquatic, though some species have tadpoles that are terrestrial.

How genetic material exchange in paramecium?

The micronucleus is the generative, or germline nucleus, containing the **genetic material** that is passed along from one generation to the next. **Paramecium** reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis

Acrosome and Its role?

Acrosome: In Eutherian mammals the **acrosome** contains digestive enzymes (including hyaluronidase and acrosin). These enzymes break down the outer membrane of the ovum, called the zona pellucida, allowing the haploid nucleus in the sperm cell to join with the haploid nucleus in the ovum.

Protostome?

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location. There are two major branches of the protostomes. **Ecdysozoa, Lophotrochozoa.**

Cadherin types?

E-cadherin P-cadherin N-cadherin Protocadherins

Role of Bicoid and Nanos in synctial specification?

The Bicoid and Nanos proteins form a coordinate system based on their ratios, such that each region of the embryo will be distinguished by a different ratio of the two proteins. Those nuclei in regions containing high amounts of Bicoid and little Nanos will be instructed to activate those genes necessary for producing the head. Those nuclei in regions that have little or no Bicoid and plenty of Nanos will be instructed to form the abdominal Structures. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates.

What are the function of zone pellicuda?

The **zona pellucida** supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and freeswimming sperm during and following fertilization.

A **blastocoel** (also spelled **blastocoele** and **blastocele**) is a fluid-filled cavity that forms in the animal hemisphere of early amphibian and echinoderm embryos, or between the epiblast and hypoblast of avian, reptilian, and mammalian blastoderm-stage embryos.

blastocoel and its **function**

A **blastocoel** is also termed the **blastocyst cavity**[2] (or *cleavage* or *segmentation cavity*) is the name given to the fluid-filled cavity of the blastula (blastocyst) that results from cleavage of the oocyte (ovum) after fertilization.[3][4] It forms during embryogenesis,[4] as what has been termed a "Third Stage" after the single-celled fertilized oocyte (zygote, ovum[1]) has divided into 16-32 cells,[3] via the process of mitosis.[5] It can be described as the first cell cavity formed as the embryo enlarges,[5] the essential precursor for the differentiated, topologically distinct, gastrula.[1]

The blastocoel is the cavity that forms within the early embryonic ball of cells. If the blastocoel can be said to have a function, I guess it would be to provide space for the invagination of one side of the hollow blastula, to form a two-layered embryo called the gastrula. That process itself is called gastrulation, and it ultimately gives rise to the three germ layers of the embryo: ectoderm, mesoderm, and endoderm

The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it

Define sex and reproduction

Sex and reproduction are two distinct and separable processes. Reproduction involves the creation of new individuals. Sex involves the combining of genes from two different individuals into new arrangements.

Diff between exon and intron

Exon: Coding region of DNA

Interon: Non Coding region of DNA

Zp3 and its role in

Thus, ZP3 is the specific glycoprotein in the mouse zona pellucida to which the sperm bind. ZP3 also initiates the acrosomal reaction after sperm have bound to it.

Transcription factor

Transcription factors are proteins that bind to enhancer or promoter regions and interact to activate or repress the transcription of a particular gene. Most transcription factors can bind to specific DNA sequences. Transcription factors have three major domains. 1. DNA-binding do-
main 2. Protein-protein interaction domain 3. Trans-activating domain

Lifecycle of calaymadomonas 5m

Let us take an example of the life cycle of Chlamydomonas. Which is usually haploid, having just one copy of each chromosome. The individuals of each species, however, are divided into two mating types: plus and minus. When a plus and a minus meet, they join their cytoplasm, and their nuclei fuse to form a diploid zygote. The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These specialized regions contain mating type-specific components that enable the cytoplasms to fuse. Following flagellar agglutination, the plus individuals initiate fusion by extending a fertilization tube. This tube contacts and fuses with a specific site on the minus individual. In evolving sexual reproduction, two important advances had to be achieved. The first was the mechanism of meiosis whereby the diploid complement of chromosomes is reduced to the haploid state. The second was a mechanism whereby the two different mating types could recognize each other.

Transcription domain

DNA-binding domain: It recognizes a particular DNA sequence.

Protein-protein interaction domain: It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

Trans-activating domain: It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this trans-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

Blastocoels .2

A fluid-filled cavity, the blastocoel, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation

Function of egg jelly. 2

Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm.

Differentiate b/w potency cell and fate cell. 3

1. **Potency** is the entire range of the cell types a particular cell can give rise to in all possible environments. For example: A cell is pluripotent if it can give rise to all different blood cell lineages. There is a decrease in potency as cells become committed.

Fate of a cell is all the different cell types, its descendants can become during normal development

- Cell potency ≥ fate.
- Potency is intrinsic property.
- Cell fate is equal to potency plus environment.
- Cell fate becomes increasingly restricted until a cell is terminally differentiated.

Differentiate b/w dyenin and the axoneme 3

The force for sperm propulsion is provided by **dynein**, a protein that is attached to the microtubules
The major motor portion of the flagellum is called the **axoneme**. It is formed by microtubules emanating from the centriole at the base of the sperm nucleus.

Detail of protostomes 5

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.
There are two major branches of the protostomes. 1. The Ecdysozoa includes those animals that molt. Its major constituent is Arthropoda, a phylum containing insects, arachnids, mites, crustaceans, and millipedes. 2. The second major group of protostomes are the Lophotrochozoa. They are characterized by a common

type of cleavage (spiral), a common larval form, and a distinctive feeding apparatus. These phyla include annelids, molluscs, and flatworms.

Step of sperm attraction with egg in simple step 5

1. The interaction of sperm and egg generally proceeds according to five basic step; 2. The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
3. The exocytosis of the acrosomal vesicle to release its enzymes.
4. The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
5. The passing of the sperm through this extracellular envelope.
6. Fusion of egg and sperm cell plasma membranes

What is protonucleus?

Pro-Nucleus is the nucleus of either gametes i.e. sperm and egg cells in the process of fertilization but before the fusion of genetic materials of both gametes.

Regulation of gene expression

The regulation of gene expression can be accomplished at several levels

Differential gene transcription: It regulates that which of the nuclear genes is transcribed into RNA.

Selective nuclear RNA processing: It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

Selective messenger RNA translation: It regulates that which of the mRNAs in the cytoplasm becomes translated into proteins.

Differential protein modification: It regulates that which proteins are allowed to remain or function in the cell.

Developmental genetics? 2

Developmental genetics is the discipline that examines how the genotype is transformed into the phenotype, and the major paradigm of developmental genetics is differential gene expression from the same nuclear repertoire.

Difference b/w Bicoid and Nanos? 2

In Drosophila, for instance, the anterior most portion of the egg contains an mRNA that encodes a protein called **Bicoid**.

The posterior most portion of the egg contains an mRNA that encodes a protein called **Nanos**.

Somit,tadpole and neural tube? 3

Somite: The mesodermal tissue adjacent to the notochord becomes segmented into somites, the precursors of the frog's back muscles, spinal cord, and dermis

Neurul Tube: Notochard is a rod of mesodermal cells in the most dorsal portion of the embryo. At this stage, the embryo is called a Neurula. The neural precursor cells elongate, stretch, and fold into the embryo forming the neural tube.

Tadpole: Somites appear as blocks of mesodermal tissue. The embryo develops a mouth and an anus, and it elongates into the typical tadpole structure.

Cumulus and crona radiata? 3

The mammalian egg is also surrounded by a layer of cells called the **cumulus**.

The innermost layer of cumulus cells, immediately adjacent to the zona pellucida, is called the **corona radiata**.

Differentiate between animal pole and vegetable pole 3

vegetal pole. :the point on the surface of an egg that is diametrically opposite to the animal pole and usually marks the center of the protoplasm containing more yolk, dividing more slowly and into larger blastomeres than that about the animal pole,

Modes of commitment 5

Three basic modes of commitment are,

Autonomous specification

Conditional specification

Syncytial specification

Autonomous specification: Specification by differential acquisition of certain cytoplasmic molecules present in the egg. Characteristic of most invertebrates. Cell type specification

Conditional specification: Characteristic of all vertebrates and few invertebrates. Specification by interactions between cells. Relative positions are important. Variable cleavages produce no invariant fate assignments to cells. o Massive cell rearrangements and migrations precede or accompany specification. o Capacity for "regulative" development: allows cells to acquire different functions.

Syncytial specification: • Characteristic of most insect classes. • Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm. Modes of commitment

• Variable cleavage produces no rigid cell fates for particular nuclei. • After cellularization, conditional specification is most often seen.

2 objectives of development 2 marks

1. It generates cellular diversity and order within each generation.
2. It ensures the continuity of life from one generation to the next.

MITF 2

(Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.

Fertilization

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Fertilization accomplishes two separate processes: Sex (the combining of genes derived from the two parents) and Reproduction (the creation of new organisms)

How we can get knowledge about gene activity in human

Knowledge of gene activity in humans can be obtained by candidate gene mapping or positional cloning.

Teratology and Evolutionary embryology

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development.

Evolutionary Embryology: The study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible

Role of Calcium and bicarbonate ions in capacitation

Calcium and bicarbonate ions may be critical in activating cAMP (Cyclic adenosine monophosphate) production and in facilitating the membrane fusion events of the acrosomal reaction

Hammerling Hypothesis:

The nucleus synthesizes a stable mRNA that lies dormant in the cytoplasm until the time of cap formation". Nucleus contains information specifying the type of cap produced (i.e., it contains the genetic information that specifies the proteins required for the production of a certain type of cap).

Fertiline and Gametogenesis:

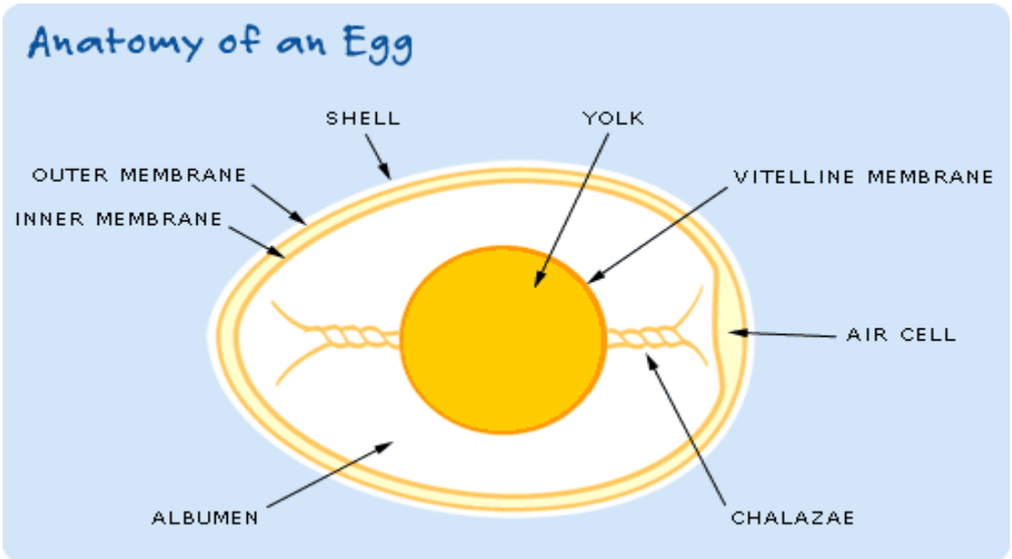
In mammals, the **fertilin** proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion.

The development of gametes, called **gametogenesis**, is usually not completed until the organism has become physically mature.

what are silencers 3 marks

▶ Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called **silencers**.

Structure of Egg Shell?



SHELL Bumpy and grainy in texture, an eggshell is covered with as many as 17,000 tiny pores. Eggshell is made almost entirely of calcium

carbonate ($CaCO_3$) crystals. It is a semipermeable membrane, which means that air and moisture can pass through its pores. The shell also has a thin outermost coating called the *bloom* or *cuticle* that helps keep out bacteria and dust.

INNER AND OUTER MEMBRANES — Lying between the eggshell and egg white, these two transparent protein membranes provide efficient defense against bacterial invasion. If you give these layers a tug, you'll find they're surprisingly strong. They're made partly of keratin, a protein that's also in human hair.

AIR CELL — An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg's larger end, and it accounts for the crater you often see at the end of a hard-cooked egg. The air cell grows larger as an egg ages.

ALBUMEN — The egg white is known as the albumen, which comes from *albus,* the Latin word for "white." Four alternating layers of thick and thin albumen contain approximately 40 different proteins, the main components of the egg white in addition to water.

CHALAZAE — Opaque ropes of egg white, the chalazae hold the yolk in the center of the egg. Like little anchors, they attach the yolk's casing to the membrane lining the eggshell. The more prominent they are, the fresher the egg.

VITELLINE MEMBRANE — The clear casing that encloses the yolk.

YOLK — The yolk contains less water and more protein than the white, some fat, and most of the vitamins and minerals of the egg. These include iron, vitamin A, vitamin D, phosphorus, calcium, thiamine, and riboflavin. The yolk is also a source of lecithin, an effective emulsifier. Yolk color ranges from just a hint of yellow to a magnificent deep orange, according to the feed and breed of the hen

First layer of egg shell makes defense system 2 marks

The Shell

The shell is the egg's outer covering, accounting for about 9 to 12% of its tota weight depending on egg size. The shell is the egg's first line of defense against bacterial contamination. This bumpy and grainy outer covering has approximately 17,000 tiny pores and is made almost entirely of calcium carbonate. It is semipermeable, which means that air and moisture can pass through its pores.

The Shell Membranes

Lying between the eggshell and albumen, these two transparent protein membranes provide efficient defense against bacterial invasion. These tough membranes are made partly of keratin, a protein also found in human hair.

1. **What is naive? 3**

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

What is microvili?3

During fertilization, these actin molecules polymerize to form long cables of actin known as **microfilaments**.

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections called **microvilli**, which may aid sperm entry into the cell.

Define zona pellucida 2 marks

Zona pellucida: The strong membrane that forms around an ovum as it develops in the ovary. The membrane remains in place during the egg's travel through the fallopian tube. To fertilize the egg, a sperm must penetrate the thinning zona pellucida. If fertilization takes place, the zona pellucida disappears, to permit implantation in the uterus

Vegetal Hemisphere?

Anatomical region that is the lower, light-colored hemisphere of the egg, as its cells contain large amounts of dense yolk proteins. The Xenopus egg is divided into two roughly equal halves: the animal and vegetal hemispheres